INCH-POUND

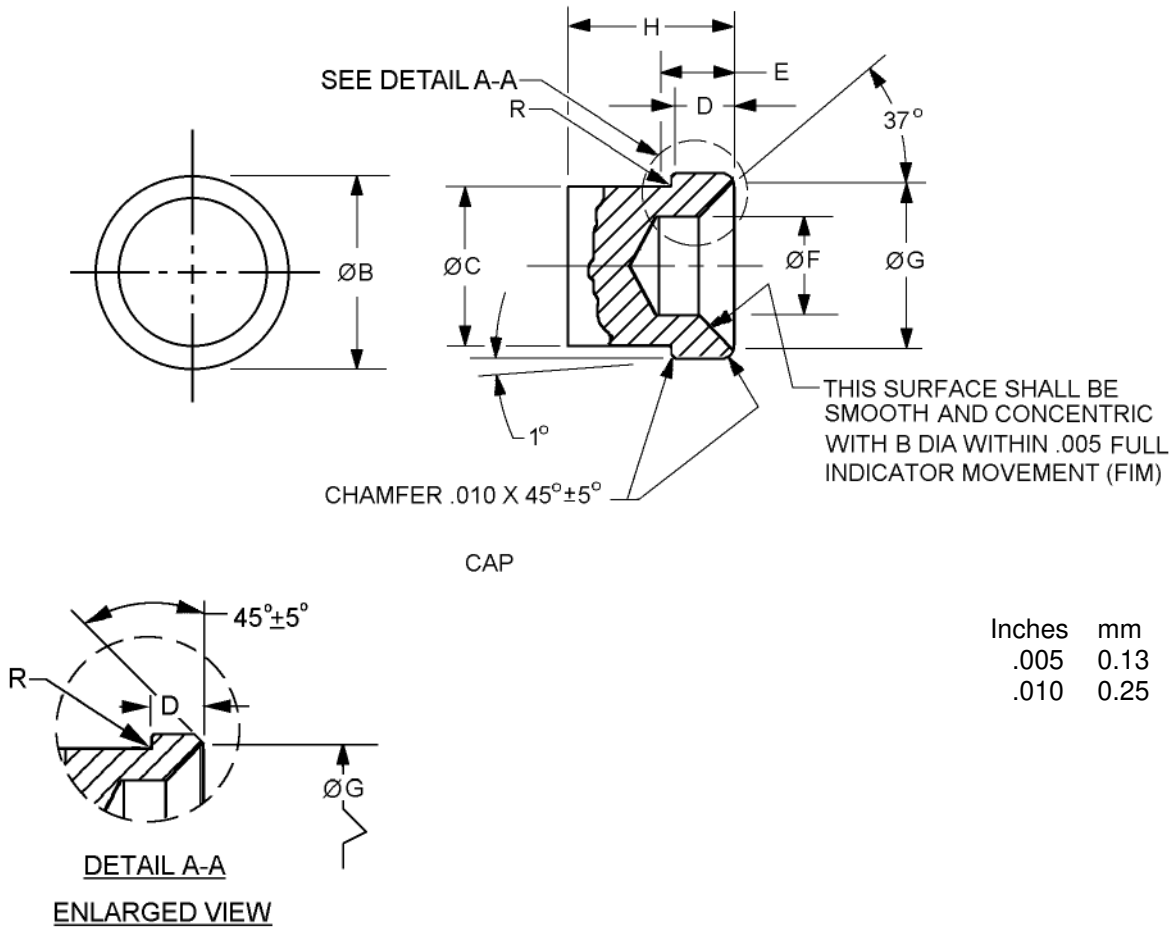
AN929 Rev 13
w/AMENDMENT 2
7 August 2014
SUPERSEDING
w/AMENDMENT 1
20 November 2013

DETAIL SPECIFICATION SHEET

# CAP INSERT, ASSEMBLY, PRESSURE SEAL, FLARED TUBE FITTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4841.

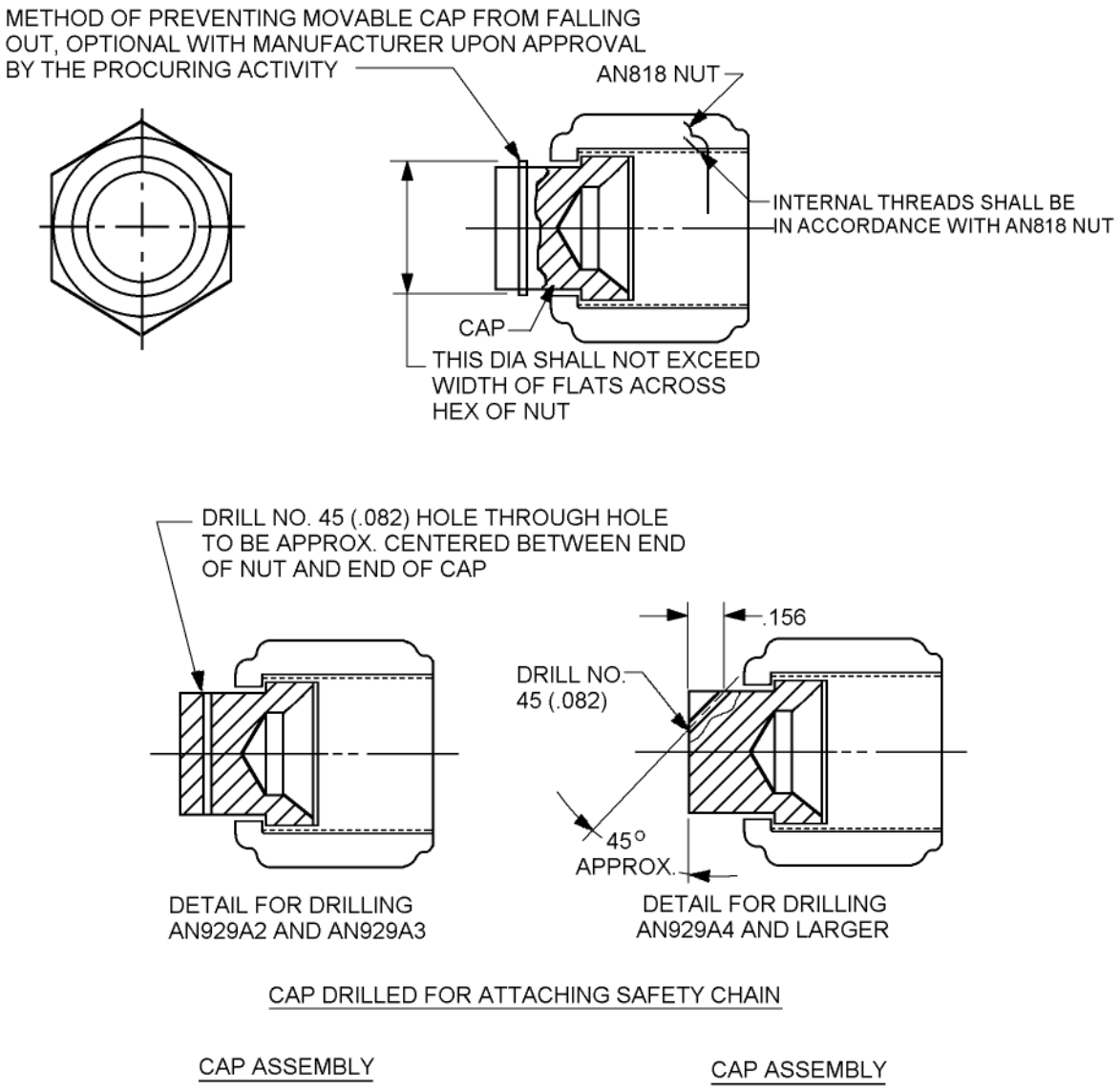



| Inches | mm |
|---|---|
| .005 | 0.13 |
| .010 | 0.25 |

FIGURE 1. Cap dimensions and configurations.



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| Inches | mm |
|---|---|
| .005 | 0.13 |
| .082 | 2.08 |
| .156 | 3.96 |

FIGURE 2. Cap assembly, dimensions and configurations.



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| Dash number | Tube OD inches (mm) | B dia inches (mm) +.000 -.003 (0.08) | C dia. inches (mm) | D inches (mm) | E inches (mm) |
|---|---|---|---|---|---|
| -2 | .125 (3.18) | .263 (6.68) | .156 (3.96) | .156 (3.96) | .125 (3.18) |
| -3 | .188 (4.78) | .325 (8.26) | .219 (5.56) | .219 (5.56) | |
| -4 | .250 (6.35) | .378 (9.60) | .281 (7.14) | .203 (5.16) | |
| -5 | .313 (7.95) | .441 (11.20) | .344 (8.78) | .219 (5.56) | .141 (3.58) |
| -6 | .375 (9.53) | .503 (12.78) | .422 (10.72) | .266 (6.76) | .156 (3.96) |
| -8 | .500 (12.70) | .683 (17.35) | .547 (13.89) | .313 (7.95) | .172 (4.37) |
| -10 | .625 (15.88) | .799 (20.29) | .672 (17.07) | .281 (7.14) | .250 (6.35) |
| -12 | .750 (19.05) | .974 (24.74) | .813 (20.65) | | .313 (7.95) |
| -16 | 1.000 (25.40) | 1.224 (31.09) | 1.063 (27.00) | .297 (7.54) | |
| -20 | 1.250 (31.75) | 1.536 (39.01) | 1.328 (33.73) | .375 (9.53) | |
| -24 | 1.500 (38.10) | 1.786 (45.36) | 1.594 (40.49) | .406 (10.31) | .375 (9.53) |
| -28 | 1.750 (44.45) | 2.161 (54.89) | 1.875 (47.63) | .500 (12.70) | |
| -32 | 2.000 (50.80) | 2.411 (61.24) | 2.141 (54.38) | .578 (14.68) | .500 (12.70) |
| -40 | 2.500 (63.50) | 2.911 (73.94) | 2.641 (67.08) | .625 (15.88) | .688 (17.48) |
| -48 | 3.000 (76.20) | 3.411 (86.64) | 3.156 (80.16) | .688 (17.48) | |

| Dash number | F dia. | G dia inches (mm) +.000 -.005 (0.13) | H inches (mm) | R radius max inches (mm) |
|---|---|---|---|---|
| -2 | .094 (2.39) | .218 (5.54) | .531 (13.49) | .005 (0.13) |
| -3 | .156 (3.96) | .250 (6.35) | | |
| -4 | .172 (4.37) | .300 (7.62) | | |
| -5 | .234 (5.94) | .360 (9.14) | .563 (14.30) | |
| -6 | .297 (7.54) | .440 (11.18) | .625 (15.88) | .010 (0.25) |
| -8 | .391 (9.93) | .575 (14.61) | .750 (19.05) | |
| -10 | .438 (11.13) | .685 (17.40) | | |
| -12 | .563 (14.30) | .885 (22.48) | .875 (22.23) | |
| -16 | .813 (20.65) | 1.103 (28.02) | | |
| -20 | 1.078 (27.38) | 1.413 (35.89) | 1.000 (25.40) | |
| -24 | 1.313 (33.35) | 1.661 (42.19) | 1.125 (28.58) | |
| -28 | 1.547 (39.29) | 2.034 (51.66) | 1.375 (34.93) | .015 (0.38) |
| -32 | 1.781 (45.24) | 2.284 (58.01) | 1.438 (36.53) | |
| -40 | 2.000 (50.80) | 2.784 (70.71) | 1.750 (44.45) | |
| -48 | 2.500 (63.50) | 3.284 (83.41) | 1.813 (46.05) | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Fittings shall be free of all burrs and slivers.
4. Unless otherwise specified tolerances for three point decimals ±.010 inch (0.25 mm).
5. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Cap assembly, dimensions and configurations. - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1 and 2.

The part is a 2-piece assembly consisting of an AN818 nut and an insert as specified herein, see figure 1.

Material: See table I herein, procurement specification SAE-AS4841.

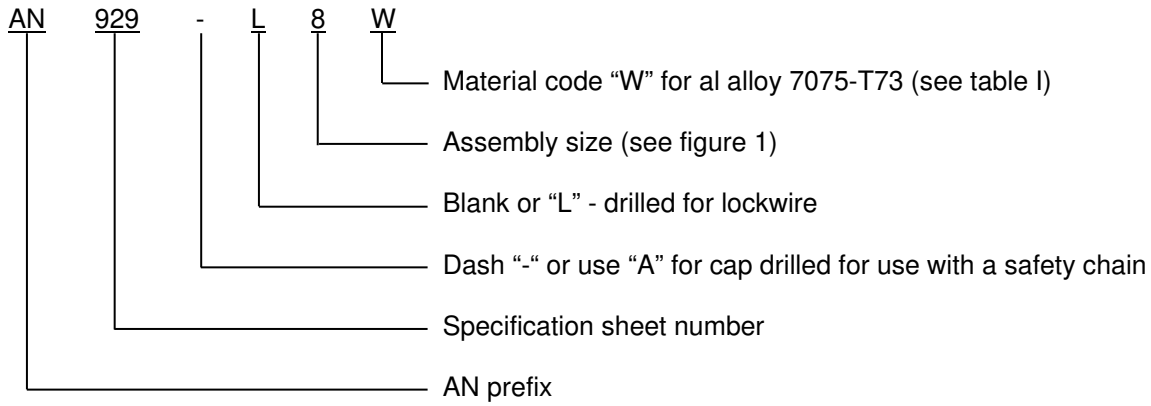
TABLE I. Material cap assembly.

| Cap insert assembly AN929 | | | | Nut AN818 2/ | | |
|---|---|---|---|---|---|---|
| Material designator | Material | Alloy | | Material designator | Material | Alloy |
| F | Steel | 4130 or 4140 | | Blank | Steel | 4130 or 4140 |
| J | CRES | 304 | | J | CRES | 304 |
| K | CRES | 316 | | K | CRES | 316 |
| S | CRES | 347 | | J or K | CRES | 304 or 316 |
| T 1/ | Titanium | 6Al-4V | | T | Titanium | 6Al-4V |
| W | Aluminum | 7075-T73 | | W | Aluminum | 7075-T73 |

1/ Not for use with oxygen systems.
2/ Material designator for manufacturers use only to determine proper AN818 nut to use for the cap insert assembly.

Finish: See SAE-AS4841.

Part or Identifying Number (PIN): The PIN consists of the AN prefix, specification sheet number, dash or “A” for cap drilled for safety chain, blank or “L” for nut drilled with lock wire, number for assembly size, and a letter for material designator.




PIN Examples:

AN929-8W identifies a cap assembly .500 inch (tube), 7075-T73 aluminum.
AN929-L8J identifies a cap assembly .500 inch (tube), nut drilled for lockwire, CRES 304.
AN929A8S identifies a cap assembly .500 inch (tube), cap drilled for attachment of safety chain, CRES 347.



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AN929AL8J identifies a cap assembly .500 inch (tube), cap drilled for use with a safety chain, and nut for lock wire, CRES 304.

Supersession data: Due to stress corrosion cracking aluminum alloys 2014 and 2024 blank designator has been replaced by aluminum alloy 7075 “W” designator see table II.

TABLE II. Supersession data.

| Cancelled AN929 PIN | | Replacement AN929 PIN | | Cancelled AN929 PIN | | Replacement AN929 PIN |
|---|---|---|---|---|---|---|
| AN929-2 | | AN929-2W | | AN929-16 | | AN929-16W |
| AN929-L2 | | AN929-L2W | | AN929-L16 | | AN929-L16W |
| AN929A2 | | AN929A2W | | AN929A16 | | AN929A16W |
| AN929AL2 | | AN929AL2W | | AN929AL16 | | AN929AL16W |
| AN929-3 | | AN929-3W | | AN929-20 | | AN929-20W |
| AN929-L3 | | AN929-L3W | | AN929-L20 | | AN929-L20W |
| AN929A3 | | AN929A3W | | AN929A20 | | AN929A20W |
| AN929AL3 | | AN929AL3W | | AN929AL20 | | AN929AL20W |
| AN929-4 | | AN929-4W | | AN929-24 | | AN929-24W |
| AN929-L4 | | AN929-L4W | | AN929-L24 | | AN929-L24W |
| AN929A4 | | AN929A4W | | AN929A24 | | AN929A24W |
| AN929AL4 | | AN929AL4W | | AN929AL24 | | AN929AL24W |
| AN929-5 | | AN929-5W | | AN929-28 | | AN929-28W |
| AN929-L5 | | AN929-L5W | | AN929-L28 | | AN929-L28W |
| AN929A5 | | AN929A5W | | AN929A28 | | AN929A28W |
| AN929AL5 | | AN929AL5W | | AN929AL28 | | AN929AL28W |
| AN929-6 | | AN929-6W | | AN929-32 | | AN929-32W |
| AN929-L6 | | AN929-L6W | | AN929-L32 | | AN929-L32W |
| AN929A6 | | AN929A6W | | AN929A32 | | AN929A32W |
| AN929AL6 | | AN929AL6W | | AN929AL32 | | AN929AL32W |
| AN929-8 | | AN929-8W | | AN929-40 | | AN929-40W |
| AN929-L8 | | AN929-L8W | | AN929-L40 | | AN929-L40W |
| AN929A8 | | AN929A8W | | AN929A40 | | AN929A40W |
| AN929AL8 | | AN929AL8W | | AN929AL40 | | AN929AL40W |
| AN929-10 | | AN929-10W | | AN929-48 | | AN929-48W |
| AN929-L10 | | AN929-L10W | | AN929-L48 | | AN929-L48W |
| AN929A10 | | AN929A10W | | AN929A48 | | AN929A48W |
| AN929AL10 | | AN929AL10W | | AN929AL48 | | AN929AL48W |
| AN929-12 | | AN929-12W | | | | |
| AN929-L12 | | AN929-L12W | | | | |
| AN929A12 | | AN929A12W | | | | |
| AN929AL12 | | AN929AL12W | | | | |

AN929 assemblies which consist of a cap and an AN818 nut shall be fastened together for shipping or stock purposes in a manner acceptable to the procuring activity.



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4841, this document references AN818.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2014-040)

Review activity:
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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